

Project Timeline Summary

Timeline	Phase	Activities	Deliverables
Week 1	Requirement Analysis	Identify AI use cases, define scope, select KPIs, determine datasets and system architecture	Project requirements document
Week 2	Data Collection & Cleaning	Gather historical energy data, handle missing values, standardize formats, validate sensor readings	Cleaned dataset
Week 3	Feature Engineering	Create ML features such as peak-hour indicators, rolling averages, trend metrics, and time-based features	ML-ready dataset
Week 4	AI Infrastructure Setup	Build FastAPI service, establish PostgreSQL integration, create training pipeline and APIs	AI service foundation
Week 5	Energy Forecasting Development	Develop consumption prediction models using historical energy data	Initial forecasting model
Week 6	Model Training & Evaluation	Train, test, and optimize forecasting models using accuracy metrics	Validated forecasting system
Week 7	Cost Prediction Development	Develop models for electricity cost and budget forecasting	Cost prediction engine
Week 8	Forecasting Dashboard Integration	Integrate forecasting outputs into dashboard cards and visualizations	Forecasting module deployed
Week 9	Consumption Pattern Analysis	Analyze historical trends and establish baseline energy behavior	Baseline consumption profiles
Week 10	Anomaly Detection Development	Implement anomaly detection models to identify unusual consumption patterns	Anomaly detection engine
Week 11	Alert System Development	Build automated alert generation and severity classification system	Intelligent alert module

Week 12	Anomaly Dashboard Integration	Integrate anomaly detection results into dashboard and reporting modules	Real-time monitoring system
Week 13	AI Assistant Foundation	Integrate LLM and create tool-calling architecture connecting chatbot with ML services	AI assistant backend
Week 14	Automated Report Generation	Generate AI-powered executive summaries, monthly reports, and recommendations	Smart report generation module
Week 15	Conversational Analytics Development	Enable natural language queries for forecasting, reports, and energy analytics	AI chatbot interface
Week 16	Testing, Optimization & Deployment	Perform system testing, optimization, documentation, and final deployment	Production-ready AI-enhanced dashboard

Milestone Overview

Month	Major Deliverable
Month 1	Data Pipeline, Feature Engineering, AI Infrastructure
Month 2	Energy Forecasting and Cost Prediction Models
Month 3	Anomaly Detection and Intelligent Alert System
Month 4	AI Chatbot, Report Generation, Testing and Deployment

Final System Capabilities After Completion

1. Energy Consumption Forecasting
2. Cost Prediction and Budget Forecasting
3. Real-Time Anomaly Detection
4. Intelligent Alert Generation
5. Automated AI Report Generation
6. Natural Language Data Queries
7. Conversational AI Assistant
8. Actionable Energy Optimization Recommendation.